

A Hierarchical Oligopolistic Execution Economy

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Abstract

This paper models an economy on the finite horizon $t \in [0, 1]$ with three hierarchical classes: Principals, Agents, and Citizens. Principals are oligopolistic decision-makers; Agents are competitive executors of Principal decisions; Citizens are price-taking residual participants. The resulting structure is a dynamic Stackelberg–Nash–Walras economy. Principals extract strategic rents, Agents are disciplined by competitive execution markets, and Citizens absorb price, welfare, fiscal, and geopolitical consequences. The paper develops the mathematical structure of the economy, characterizes equilibrium, derives welfare distortions, introduces political and warfare-economic extensions, and proposes statistical and AI/ML verification algorithms.

The paper ends with “The End”

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1 Introduction

Consider an economy existing over the finite interval

$$t \in [0, T], \quad T = 1.$$

The population is partitioned into three classes:

$$\mathcal{P} = \{1, \dots, N\}, \quad \mathcal{A} = \{1, \dots, M\}, \quad \mathcal{C} = \{1, \dots, L\}.$$

The hierarchy is

$$\boxed{\text{Principals} \longrightarrow \text{Agents} \longrightarrow \text{Citizens.}}$$

Principals are oligopolists. They decide. Agents are competitors. They execute. Citizens are neither Principals nor Agents. They are price-takers.

The central object is a hierarchical economy in which strategic authority is concentrated, operational execution is competitive, and social consequences are distributed across passive price-taking citizens.

2 Basic Structure of the Economy

2.1 Classes

Class	Economic role	Strategic status
Principals	Decide, own, command, invest	Strategic oligopolists
Agents	Execute, manage, compete	Competitive implementers
Citizens	Consume, work, pay prices	Price-takers

Table 1: Hierarchy of the economy

Let aggregate output be

$$X(t) = \sum_{i \in \mathcal{P}} x_i(t),$$

and let price be $p(t)$. For simplicity, begin with one good. The multi-good version replaces scalars by vectors.

3 Citizens

Each Citizen $c \in \mathcal{C}$ chooses consumption $q_c(t)$ and labor $\ell_c(t)$ to solve

$$\max_{q_c, \ell_c} \int_0^1 e^{-\rho t} u_c(q_c(t), 1 - \ell_c(t)) dt$$

subject to

$$p(t)q_c(t) \leq w_c(t)\ell_c(t) + \omega_c(t).$$

The first-order condition is

$$\frac{\partial u_c / \partial q_c}{\partial u_c / \partial (1 - \ell_c)} = \frac{p(t)}{w_c(t)}.$$

Aggregate demand is

$$D(p(t), Y(t)) = \sum_{c \in \mathcal{C}} q_c(p(t), y_c(t)).$$

Citizens are price-takers:

$$\frac{\partial p(t)}{\partial q_c(t)} = 0.$$

4 Agents

Agents execute Principal decisions. Agent $j \in \mathcal{A}$ supplies effort $e_j(t)$ and receives wage or contract payment $w_j(t)$.

The Agent solves

$$\max_{e_j} \int_0^1 e^{-\delta t} [w_j(t) - \psi_j(e_j(t))] dt.$$

The participation constraint is

$$V_j \geq \bar{V}_j,$$

where

$$V_j = \int_0^1 e^{-\delta t} [w_j(t) - \psi_j(e_j(t))] dt.$$

The incentive compatibility condition is

$$e_j^*(t) \in \arg \max_{e_j(t)} [w_j(t) - \psi_j(e_j(t))].$$

If effort is contractible, then

$$w_j(t) = \psi_j(e_j(t)) + \bar{v}_j(t).$$

If effort is not contractible, moral hazard appears.

5 Principals

Principal $i \in \mathcal{P}$ chooses strategic control

$$s_i(t) \in S_i,$$

where $s_i(t)$ may include quantity, price, capital, lobbying, military expenditure, information control, contract design, or institutional capture.

Profit is

$$\Pi_i = \int_0^1 e^{-rt} [p(t)x_i(t) - C_i(x_i(t), a_i(t)) - W_i(t) - G_i(t)] dt.$$

Here:

$$C_i = \text{production cost}, \quad W_i = \text{payments to Agents}, \quad G_i = \text{political/geopolitical expenditure}.$$

The Principal's problem is

$$\max_{s_i(\cdot)} \Pi_i(s_i, s_{-i})$$

subject to demand, execution, and institutional constraints.

6 Market Clearing

Goods-market clearing:

$$\sum_{i \in \mathcal{P}} x_i(t) = D(p(t), Y(t)).$$

Execution-market clearing:

$$\sum_{i \in \mathcal{P}} a_i(t) = \sum_{j \in \mathcal{A}} e_j(t).$$

Budget feasibility:

$$\sum_{c \in \mathcal{C}} p(t) q_c(t) \leq \sum_{c \in \mathcal{C}} y_c(t).$$

7 Equilibrium

Definition 1 (Hierarchical Oligopolistic Execution Equilibrium). *A hierarchical oligopolistic execution equilibrium is a collection*

$$(s_i^*, x_i^*, e_j^*, q_c^*, p^*, w^*)_{t \in [0,1]}$$

such that:

(i) each Principal chooses a best response:

$$s_i^* \in \arg \max_{s_i \in S_i} \Pi_i(s_i, s_{-i}^*);$$

(ii) each Agent optimally executes:

$$e_j^* \in \arg \max_{e_j} [w_j - \psi_j(e_j)];$$

(iii) each Citizen optimizes competitively:

$$q_c^* \in \arg \max_{q_c} u_c(q_c, 1 - \ell_c);$$

(iv) goods markets clear:

$$\sum_i x_i^*(t) = \sum_c q_c^*(t);$$

(v) execution markets clear:

$$\sum_i a_i^*(t) = \sum_j e_j^*(t).$$

8 Cournot Characterization

Assume inverse demand

$$p(t) = A - BX(t), \quad X(t) = \sum_{i=1}^N x_i(t),$$

and constant marginal cost

$$C_i(x_i) = cx_i.$$

Principal i solves

$$\max_{x_i} \left(A - B \sum_{k=1}^N x_k \right) x_i - cx_i.$$

The first-order condition is

$$A - c - BX - Bx_i = 0.$$

In symmetric equilibrium, $x_i = x$, so $X = Nx$. Therefore

$$A - c - BNx - Bx = 0,$$

hence

$$x^* = \frac{A - c}{B(N + 1)}.$$

Aggregate output is

$$X^* = \frac{N(A - c)}{B(N + 1)}.$$

Price is

$$p^* = A - BX^* = A - \frac{N(A - c)}{N + 1} = \frac{A + Nc}{N + 1}.$$

Profit per Principal is

$$\pi_i^* = (p^* - c)x^* = \left(\frac{A - c}{N + 1} \right) \left(\frac{A - c}{B(N + 1)} \right) = \frac{(A - c)^2}{B(N + 1)^2}.$$

Proposition 1 (Oligopoly distortion). *For finite N , the hierarchical economy produces less and charges more than the competitive economy.*

Proof. The competitive price is $p^C = c$. Competitive quantity satisfies

$$A - BX^C = c,$$

so

$$X^C = \frac{A - c}{B}.$$

Since

$$X^* = \frac{N(A - c)}{B(N + 1)} < \frac{A - c}{B} = X^C,$$

output is lower. Also

$$p^* = \frac{A + Nc}{N + 1} > c = p^C.$$

Thus oligopoly produces less and charges more. □

9 Surplus Decomposition

Total surplus is

$$S(t) = CS(t) + PS(t) + AS(t),$$

where

$$CS = \text{Citizen surplus}, \quad PS = \text{Principal surplus}, \quad AS = \text{Agent surplus}.$$

Because Agents are competitive,

$$AS(t) \approx 0$$

above reservation value.

Principal surplus is

$$PS(t) = \sum_i \pi_i(t).$$

Citizen surplus under linear demand is

$$CS(t) = \frac{1}{2}(A - p(t))X(t).$$

Under Cournot:

$$CS^* = \frac{1}{2} \left(A - \frac{A + Nc}{N + 1} \right) \left(\frac{N(A - c)}{B(N + 1)} \right).$$

Thus

$$CS^* = \frac{N(A - c)^2}{2B(N + 1)^2}.$$

Principal surplus is

$$PS^* = N \frac{(A - c)^2}{B(N + 1)^2}.$$

Therefore,

$$\frac{PS^*}{CS^*} = 2.$$

In the symmetric linear Cournot case, Principals receive twice the Citizen surplus.

10 Political Economy Extension

Let Principal i spend $g_i(t)$ on political influence. Institutional advantage is

$$\theta_i(t) = \Theta_i(g_i(t), g_{-i}(t)).$$

Effective cost becomes

$$\tilde{C}_i = C_i - \theta_i.$$

Profit becomes

$$\Pi_i = \int_0^1 e^{-rt} [p(t)x_i(t) - C_i(x_i(t)) + \theta_i(t) - g_i(t)] dt.$$

The first-order condition for political spending is

$$\frac{\partial \theta_i}{\partial g_i} = 1.$$

Thus lobbying or institutional capture continues until marginal political benefit equals marginal political cost.

11 Geopolitical and Warfare-Economic Extension

Let $m_i(t)$ denote coercive, military, or security expenditure. Let the strategic survival probability of Principal i be

$$\sigma_i(t) = \sigma_i(m_i(t), m_{-i}(t), z(t)),$$

where $z(t)$ is the geopolitical state.

The expected payoff becomes

$$\Pi_i = \int_0^1 e^{-rt} \sigma_i(t) [p(t)x_i(t) - C_i(x_i(t)) - W_i(t)] - m_i(t) dt.$$

The warfare-economic first-order condition is

$$\frac{\partial \sigma_i}{\partial m_i} [px_i - C_i - W_i] = 1.$$

Thus military or coercive investment rises with the value of protected rents.

12 Dynamic State Variables

Let the state vector be

$$z(t) = \begin{bmatrix} K(t) \\ H(t) \\ I(t) \\ M(t) \\ R(t) \end{bmatrix},$$

where

K = capital, H = human capital, I = institutional quality, M = military capacity, R = rent concentration

The law of motion is

$$\dot{z}(t) = F(z(t), s(t), e(t), q(t)).$$

A simple system is

$$\begin{aligned} \dot{K} &= s_K Y - \delta_K K, \\ \dot{H} &= s_H Y - \delta_H H, \\ \dot{I} &= \alpha_I I_0 - \beta_I R, \\ \dot{M} &= s_M Y - \delta_M M, \\ \dot{R} &= \gamma_R PS - \eta_R CS. \end{aligned}$$

This system captures the possibility that rent concentration weakens institutions, while surplus-sharing improves social stability.

13 Welfare Functional

A social planner evaluates

$$W = \int_0^1 e^{-\rho t} [\lambda_C U_C(t) + \lambda_A U_A(t) + \lambda_P U_P(t) - \Omega(R(t)) - \Phi(M(t))] dt.$$

Here $\Omega(R)$ is the social cost of rent concentration and $\Phi(M)$ is the social cost of militarization.

The first-best problem is

$$\max_{x, e, q, g, m} W.$$

The decentralized hierarchical equilibrium generally fails to solve this problem.

Theorem 1 (Hierarchical inefficiency). *If Principals possess market power and Citizens are price-takers, then the decentralized hierarchical equilibrium is generically not Pareto efficient.*

Proof. Under market power, Principals set marginal revenue equal to marginal cost:

$$MR = MC.$$

Efficiency requires

$$p = MC.$$

Since under downward-sloping demand

$$MR < p,$$

the oligopolistic equilibrium satisfies

$$p > MC.$$

Therefore quantity is below the efficient level. Hence the allocation is not Pareto efficient. $\square$

14 Statistical Identification

Suppose we observe panel data

$$\{p_{it}, q_{it}, w_{jt}, e_{jt}, y_{ct}\}.$$

A reduced-form empirical model is

$$p_t = \alpha + \beta HHI_t + \gamma Z_t + \varepsilon_t,$$

where

$$HHI_t = \sum_{i=1}^N s_{it}^2$$

is the Herfindahl–Hirschman concentration index.

Expected signs:

$$\beta > 0,$$

because higher concentration increases price.

Agent discipline can be estimated from

$$w_{jt} = \alpha_A + \beta_A e_{jt} + \gamma_A u_t + \varepsilon_{jt},$$

where u_t is unemployment or outside-option pressure.

Citizen welfare can be estimated from

$$CS_t = \alpha_C - \beta_C p_t + \gamma_C y_t + \varepsilon_t.$$

Equation	Variable of interest	Interpretation
$p_t = \alpha + \beta HHI_t + \varepsilon_t$	β	Oligopoly price effect
$w_{jt} = \alpha_A + \beta_A e_{jt} + \varepsilon_{jt}$	β_A	Execution reward gradient
$CS_t = \alpha_C - \beta_C p_t + \varepsilon_t$	β_C	Citizen price burden
$R_t = \alpha_R + \beta_R PS_t - \gamma_R CS_t + \varepsilon_t$	β_R, γ_R	Rent concentration dynamics

Table 2: Empirical identification equations

15 AI/ML Verification

The economy can be studied using simulation, reinforcement learning, and structural estimation.

15.1 Supervised Learning

Predict prices:

$$\hat{p}_t = f_\theta(HHI_t, X_t, W_t, G_t, M_t, Z_t).$$

Loss:

$$\mathcal{L}(\theta) = \frac{1}{T} \sum_t (p_t - \hat{p}_t)^2.$$

15.2 Reinforcement Learning

Each Principal is an RL agent with state

$$s_t = (p_t, X_t, K_t, I_t, R_t, M_t),$$

action

$$a_{it} = (x_{it}, g_{it}, m_{it}, w_{it}),$$

and reward

$$r_{it} = p_t x_{it} - C_i(x_{it}) - W_i(t) - G_i(t) - M_i(t).$$

The Bellman equation is

$$V_i(s) = \max_{a_i} [r_i(s, a_i, a_{-i}) + \beta \mathbb{E} V_i(s')].$$

15.3 Multi-Agent Learning

A multi-agent Q-learning update is

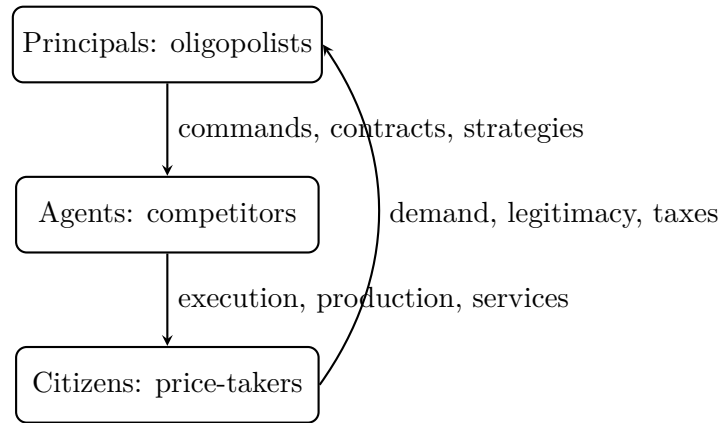
$$Q_i(s_t, a_t) \leftarrow (1 - \alpha)Q_i(s_t, a_t) + \alpha \left[r_{it} + \beta \max_{a'_i} Q_i(s_{t+1}, a'_i, a'_{-i}) \right].$$

16 Algorithm

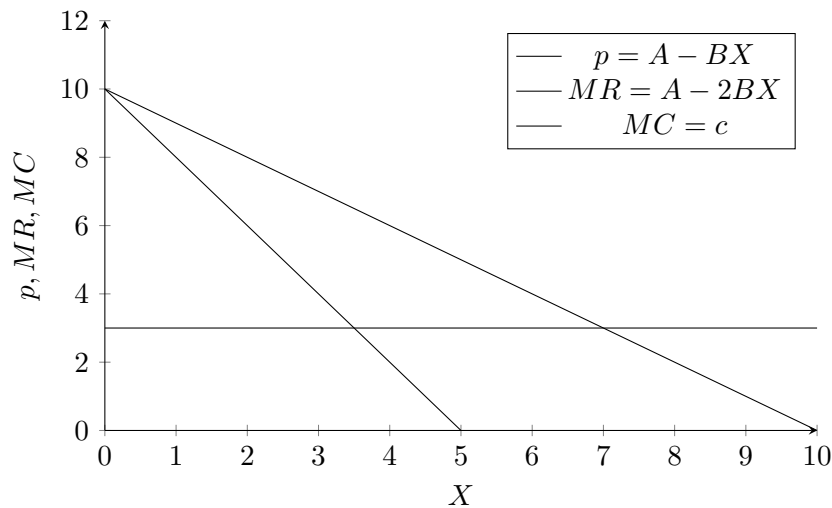
Algorithm: Simulation of a Hierarchical Oligopolistic Execution Economy

1. Initialize $N, M, L, K_0, H_0, I_0, M_0, R_0$.
 2. Initialize Principal strategies s_i^0 .
 3. For each time step t :
 - 3.1 Principals choose quantities, contracts, political spending, and security spending.
 - 3.2 Agents observe contracts and choose effort.
 - 3.3 Citizens observe prices and choose consumption.
 - 3.4 Markets clear and prices update.
 - 3.5 Profits, wages, surplus, and welfare are computed.
 - 3.6 State variables evolve according to $\dot{z} = F(z, s, e, q)$.
 4. Iterate until $t = 1$.
 5. Estimate equilibrium distortions: price markup, rent concentration, welfare loss.
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17 Vector Graphic of the Hierarchy



18 Oligopoly Price Distortion Diagram



19 Interpretation

The economy has three core mechanisms.

First, Principals possess strategic scarcity. They choose output, contracts, political investments, and coercive capacity.

Second, Agents are disciplined through competition. Their rents are bounded by outside options.

Third, Citizens are price-takers. They experience prices, taxes, inflation, shortages, war risk, and institutional decline without directly controlling strategic decisions.

Thus the economy may be summarized as

strategic concentration + competitive execution + diffuse social absorption.

20 Main Results

Result	Mathematical condition	Consequence
Oligopoly pricing	$p > MC$	Citizen welfare loss
Agent competition	$V_j = \bar{V}_j$	Low execution rents
Political capture	$\partial\theta_i/\partial g_i = 1$	Institutional distortion
Militarization	$\partial\sigma_i/\partial m_i \cdot \pi_i = 1$	Security-rent coupling
Rent concentration	$\dot{R} = \gamma_R PS - \eta_R CS$	Hierarchical persistence

Table 3: Main characterization results

21 Conclusion

A hierarchical oligopolistic execution economy is a finite-horizon dynamic economy in which Principals decide, Agents execute, and Citizens absorb the consequences. Its equilibrium combines oligopoly, contract theory, political economy, and price-taking consumption.

The defining distortion is not merely monopoly power, but hierarchical asymmetry:

decision power $\neq$ execution burden $\neq$ welfare exposure.

This asymmetry generates persistent rent concentration, political capture, and possible militarization when protected rents are sufficiently valuable.

References

- [1] Cournot, A. A. *Researches into the Mathematical Principles of the Theory of Wealth*. 1838.
- [2] Tirole, J. *The Theory of Industrial Organization*. MIT Press, 1988.
- [3] Mas-Colell, A., Whinston, M. D., and Green, J. R. *Microeconomic Theory*. Oxford University Press, 1995.
- [4] Myerson, R. B. *Game Theory: Analysis of Conflict*. Harvard University Press, 1991.
- [5] Grossman, S. J. and Hart, O. D. “The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration.” *Journal of Political Economy*, 1986.
- [6] Hart, O. and Moore, J. “Property Rights and the Nature of the Firm.” *Journal of Political Economy*, 1990.
- [7] Olson, M. *The Logic of Collective Action*. Harvard University Press, 1965.
- [8] North, D. C. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, 1990.
- [9] Schelling, T. C. *Arms and Influence*. Yale University Press, 1966.

Glossary

Principal

A strategic oligopolist who owns, commands, decides, or controls key economic variables.

Agent

A competitive executor who implements Principal decisions under contracts.

Citizen

A price-taking participant who consumes, works, pays taxes, and bears equilibrium consequences.

Oligopoly

A market structure with a finite number of strategic firms or decision-makers.

Execution market

The competitive market in which Agents supply effort or implementation capacity.

Rent concentration

The accumulation of surplus by Principals due to strategic scarcity or market power.

Political capture

The use of resources to influence institutions in favor of Principals.

Warfare economics

The study of coercive, military, and security expenditure as instruments for protecting rents.

Hierarchical equilibrium

An equilibrium in which different classes optimize under asymmetric strategic power.

Citizen surplus

Welfare accruing to price-taking Citizens from consumption.

The End